

Fault-Onset Labeling Bounds Reported Accuracy in the Tennessee Eastman Benchmark: A Leakage-Controlled Study of Six Classical Classifiers

Abstract

Fault diagnosis in the Tennessee Eastman Process (TEP) is a standard benchmark for data-driven process monitoring, yet published results are difficult to compare. We present a leakage-controlled benchmark of six classical classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear support vector machine and XGBoost. Data were partitioned by complete simulation runs, standardization was fitted on training data only, hyperparameters were selected on a separate validation partition, variability was measured over five seeds, and the frozen model was evaluated once on the complete official test partition of 10.08 million observations. XGBoost ranked first in repeated validation (macro-F1 0.7683 ± 0.0017) and reached a test macro-F1 of 0.7200, accuracy of 0.6754 and MCC of 0.6621 under the labels distributed with the dataset.

These figures are bounded by the benchmark itself. Each fault is injected 160 samples into every 960-sample test run, so literal labeling caps the recall of every fault class at 0.8333; faults 6 and 7 attain 0.8333 and 0.8332, the ceiling itself. Rescoring the same frozen model over the post-onset segment, without refitting, raises macro-F1 to 0.8054, accuracy to 0.7949 and MCC to 0.7852, and the apparent validation-to-test generalization gap of +0.0483 becomes -0.0125 under a symmetric comparison. Across seven classifiers the relative accuracy gain lies between 0.142 and 0.150 against a structural bound of 0.1587, so the effect is a property of the labeling convention rather than of any estimator. Normal operation and faults 3, 9 and 15 remain mutually confounded after the correction, consistent with their known weak observability.

Keywords: Tennessee Eastman Process, fault diagnosis, machine learning, process monitoring, reproducible benchmark, fault-onset labeling

31 1. Introduction

32 Industrial processes operate through interacting physical, chemical, and
33 control mechanisms. A local disturbance can propagate through feedback
34 loops, alter several measured variables, and eventually compromise product
35 quality, equipment integrity, safety, or environmental performance. Fault
36 detection and diagnosis (FDD) therefore constitute central elements of mod-
37 ern process monitoring [1, 2, 3]. Detection determines whether the process
38 has departed from acceptable operation, while diagnosis seeks to identify the
39 underlying condition so that operators or automated systems can respond
40 appropriately.

41 Model-based diagnosis can be effective when reliable first-principles mod-
42 els are available. However, plant-wide systems are frequently nonlinear, high-
43 dimensional, stochastic, and only partially observed. These properties have
44 motivated data-driven monitoring, in which statistical or machine-learning
45 models infer diagnostic boundaries from historical measurements [4, 2]. Clas-
46 sical multivariate methods such as principal-component analysis (PCA), par-
47 tial least squares (PLS), and Fisher discriminant analysis established impor-
48 tant foundations for process monitoring [5]. More recent studies increasingly
49 employ tree ensembles, support vector machines, neural networks, and se-
50 quence models to represent nonlinear relationships and dynamic behavior.

51 The Tennessee Eastman Process (TEP), originally introduced by Downs
52 and Vogel [6], remains one of the most widely used benchmarks in this field.
53 It represents a reactor–separator–recycle chemical process with 41 measured
54 variables, 11 manipulated variables, and a diverse set of disturbances. Its
55 continued value arises from the coexistence of relatively distinctive faults
56 and difficult conditions whose trajectories overlap normal operation. The
57 expanded simulation data published by Rieth et al. [7] provide hundreds
58 of independent runs per condition and non-overlapping random seeds for
59 development and testing, enabling large-scale and reproducible evaluation.

60 Despite extensive use of the TEP, reported results are often difficult to
61 compare. Studies may split temporally adjacent observations rather than
62 complete runs, tune models on the test set, report only accuracy, or omit
63 class-level behavior. Treating autocorrelated observations as independent
64 experimental replicates can produce optimistic estimates, while a high global
65 score can hide failures in specific operating conditions. These issues are es-
66 pecially relevant when the normal state is confused with a fault, because false
67 alarms can reduce operator confidence and generate unnecessary interven-

tions.

This study addresses these limitations through a leakage-controlled benchmark of six classical supervised classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear support vector machine (SVM), and XGBoost. The study asks five questions:

1. Which classical model provides the strongest macro-averaged diagnostic performance under repeated validation?
2. Does the selected model maintain its performance on a complete, independent official test partition?
3. Which operating conditions account for the remaining diagnostic errors?
4. How much of the reported error is attributable to the fault-onset labeling convention rather than to the classifier?
5. Which model offers the best compromise between diagnostic performance and computational cost?

The main contribution is an auditable protocol that separates model selection from final testing, preserves complete simulation runs, quantifies variability over five seeds, reports confidence intervals, and examines precision, recall, F1, and confusion patterns for all 21 classes. Rather than presenting a single headline score, the analysis identifies a compact ambiguity group that defines the practical limit of the selected classifier.

A second contribution is methodological and, in our view, of wider consequence for the TEP literature. The expanded dataset injects each fault after a fixed warm-up interval, so a fraction of every faulty run is nominally normal yet carries the fault label. This fraction differs between development runs (4%) and official test runs (16.7%). We quantify how much of the reported error and of the apparent generalization gap is produced by this convention alone, and we report every headline metric under both the literal labeling and a post-onset labeling. To our knowledge this decomposition is not routinely reported, which may account for part of the dispersion among published TEP results.

2. Related Work

2.1. Data-driven process monitoring

Data-driven FDD spans latent-variable modeling, classification, clustering, change detection, and temporal modeling. The review by Venkatasubramanian et al. [1] established a broad taxonomy covering quantitative models,

104 qualitative models, and process-history-based methods. Chiang et al. [5, 4]
 105 showed how PCA, discriminant PLS, and Fisher discriminant analysis could
 106 support chemical-process diagnosis. Later reviews emphasized that industrial
 107 data may be nonlinear, dynamic, nonstationary, multimodal, and autocorre-
 108 lated [2, 3]. These properties motivate flexible nonlinear classifiers, but also
 109 demand careful validation.

110 Multivariate statistical monitoring of the TEP has a long history. Raich
 111 and Çinar combined statistical process monitoring with disturbance diagnosis
 112 in multivariable continuous processes [20], and multiscale extensions using
 113 wavelet decompositions were introduced to separate phenomena occurring
 114 at different time scales [22]. A broader survey of plant-wide disturbance
 115 detection is given by Thornhill and Horch [25].

116 A substantial line of work addresses the serial correlation of process data
 117 directly. Dynamic PCA augments the data matrix with time-lagged variables
 118 to capture time-varying structure [35]; canonical variate analysis models the
 119 dynamics explicitly and was compared against PCA and DPCA on the TEP
 120 simulator by Russell et al. [36], later extended with kernel density estimation
 121 for nonlinear dynamics [37]. These methods are relevant here for a reason
 122 beyond their performance: they treat the temporal structure of a run as
 123 the object of analysis, whereas an observation-level classifier treats each row
 124 independently. The labeling question examined in Section 3.2 arises precisely
 125 at that boundary.

126 2.2. Classical machine learning for TEP diagnosis

127 Decision trees recursively partition the feature space into regions associ-
 128 ated with class labels [8]. Random forests reduce the instability of individual
 129 trees by combining bootstrap samples and randomized feature subsets [9].
 130 Gradient boosting instead constructs an additive ensemble in which succes-
 131 sive learners correct residual errors [10]. XGBoost extends this principle with
 132 regularized objectives, efficient histogram-oriented computation, column sub-
 133 sampling, and systems-level optimizations [11]. These characteristics make
 134 tree ensembles attractive for TEP data, where variables interact nonlinearly.

135 Linear models provide important controls. Multinomial logistic regression
 136 tests whether approximately linear decision surfaces are sufficient, whereas
 137 linear SVMs maximize class separation through margin-based learning [12].
 138 Their inclusion helps determine whether the added complexity of nonlinear
 139 ensembles is justified. All implementations in this study were based on scikit-
 140 learn [13], except XGBoost, which used its reference implementation [11].

Support vector machines have been applied to the TEP in several forms. Chiang et al. combined Fisher discriminant analysis with support vector machines for fault diagnosis [23]; Kulkarni et al. incorporated process knowledge into the kernel to detect TEP faults [24]; and kernel extensions of PCA and independent component analysis were compared against SVM on the same benchmark [26]. These studies use kernels that the present work could not afford at the sample size considered here, which is the reason the linear surrogate of Section 3.4 should be read as a lower bound rather than as the best attainable margin classifier.

Deep architectures have been applied extensively to the same benchmark. Chadha and Schwung compared feedforward and convolutional networks for TEP fault detection [31]; hybrid designs combining support vector machines with latent-variable preprocessing [34], hierarchical LSTM structures [32], and metaheuristically optimized deep models [33] have all reported strong results. Most directly comparable to the present study, Lomov et al. [30] evaluated recurrent, attention-based and temporal convolutional architectures on the same expanded TEP dataset used here. The present work does not propose a competing architecture; it examines a property of the evaluation protocol that applies to all of them.

2.3. *Evaluation protocols and leakage*

Concern about evaluation protocol is not specific to process monitoring. In a survey across disciplines that have adopted machine learning, Kapoor and Narayanan identified leakage in seventeen fields, affecting at least 294 publications and in several cases producing conclusions that did not survive correction [29]. Their taxonomy distinguishes eight mechanisms, two of which bear directly on the present benchmark: temporal leakage, where information from the future informs a prediction about the past, and non-independence between training and evaluation samples, where records that share a common origin are distributed across partitions and treated as independent replicates.

Both are latent in observation-level TEP classification. Consecutive samples within a simulation run are strongly autocorrelated, so a split performed at the observation level places near-duplicate records on both sides of the partition. The run-level protocol of Section 3.3 is designed to avoid this. The labeling question analyzed in this paper is related but distinct: it concerns not how records are divided, but whether the label attached to a record describes the state of the plant at that instant.

177 *2.4. Evaluation beyond accuracy*

178 Accuracy can be misleading when performance varies substantially among
179 classes. Macro-F1 gives equal importance to every class by averaging class-
180 specific F1 scores, while weighted-F1 weights them by support. Balanced
181 accuracy averages class recalls. MCC summarizes agreement between pre-
182 dictions and true labels and remains informative for multiclass problems [14].
183 Confusion matrices are indispensable because they reveal whether errors are
184 diffuse or concentrated in specific class pairs. The present study therefore
185 uses macro-F1 as the primary model-selection criterion and treats the other
186 metrics as complementary evidence.

187 *2.5. Reported treatment of the fault-onset window*

188 The analysis reported in this paper is only consequential if the treat-
189 ment of the onset window varies, or goes unstated, across published work
190 on the same benchmark. We therefore examined the data-description and
191 experimental-protocol sections of studies for which the full text was avail-
192 able, recording the diagnostic task, the stated treatment of the pre-onset
193 segment, and the number of test samples scored against the fault label. The
194 purpose is illustrative rather than exhaustive. Three studies on the same
195 simulator adopt three different conventions, which is sufficient to establish
196 that the choice is not standardized; Table 1 reports them.

197 The question applies to both distributions of the benchmark. In the
198 original data of Downs and Vogel as distributed by Chiang et al. [6, 4],
199 each condition is simulated for 24 hours in training and 48 hours in testing
200 at a three-minute interval, giving 480 and 960 samples, with the fault in-
201 jected after one hour and eight hours respectively. The pre-onset fractions
202 are therefore 4.2% and 16.7%, essentially identical to those of the expanded
203 dataset analyzed here. The artifact is thus a property of the benchmark
204 design rather than of any particular redistribution of it, and the results of
205 Section 4.4 bound what can be reported on either version.

206 Yin et al. [27] address the window directly. Their data description states
207 that fault information emerges eight hours after start-up, so that the first 160
208 test samples appear normal while the remaining 800 are genuinely faulty, and
209 their discussion of the classification output is explicit about the consequence:
210 the target label is -1 over the first 160 samples and $+1$ from the 161st
211 onward. The pre-onset segment is thus relabeled as normal rather than
212 scored against the fault, and their fault-detection rate is computed over 800
213 samples.

Table 1: Treatment of the pre-onset window in observation-level studies on the Tennessee Eastman benchmark. Three studies on the same simulator adopt three distinct conventions: relabeling the pre-onset segment as normal, retaining it under the fault label, and not addressing it.

Study	Task	Methods	Pre-onset handling	Scored as fault
Yin et al. [27]	binary detection	SVM, PLS	stated, relabeled normal	800
Sun et al. [28]	detection + identif.	Bayesian RNN	stated, retained	960
Agarwal et al. [32]	21-class diagnosis	LSTM-SAE	not mentioned	960
This work	21-class diagnosis	six classical	stated, both protocols	960 / 800

Sun et al. [28] state the same structure – the simulator runs 160 samples in the normal state before the disturbance is introduced, then continues for 800 further samples – but evaluate over all 960. Awareness of the injection instant is therefore not sufficient by itself; what determines comparability is what is done with the segment.

Agarwal et al. [32] describe their data as 1440 samples per class, of which the first 480 are used for calibration and the remaining 960 for testing, and identify the source distribution explicitly. The instant of fault injection is not mentioned anywhere in the data description or the experimental protocol. The consequence is not incidental to that study’s argument: its stated objective is to improve the diagnosis of the incipient faults 3, 9 and 15, to which end pseudo-random binary signals are actively injected into the plant. Under literal labeling, part of the difficulty that motivates such an intervention is contributed by observations in which no fault is yet present.

The pattern across the three is not one of carelessness, and it admits a structural explanation. In binary detection the pre-onset segment has a natural destination: the normal class already exists in the problem, so relabeling those observations is the obvious treatment, and the false-alarm rate is in fact defined over precisely that segment. The convention is therefore built into the evaluation. In multiclass diagnosis the same move is no longer neutral, because assigning pre-onset observations of every faulty run to class 0 alters the prior of the normal condition and, in the present data, would add 1.6 million observations to a class that already competes with faults 3, 9 and 15. Faced with that, the simplest course is to label each run by its fault number throughout, and the distinction disappears without ever being posed as a choice. What is lost in the transition from detection to diagnosis is not information about the data but a convention for handling it.

241 The practical consequence is that reported figures are not commensurable
 242 across the three treatments. For a single frozen model the difference between
 243 literal and post-onset labeling is 0.0854 in macro-F1 and 0.1231 in MCC (Sec-
 244 tion 4.4), larger than the margins that typically separate competing methods
 245 on this benchmark. We do not claim a systematic review here; the point is
 246 that three conventions coexist, that the choice is seldom identified as such,
 247 and that it should be reported alongside the unit of partitioning.

248 **[TBD: Optional: extend Table 1 with further studies as their**
 249 **full texts are consulted, in particular Yin et al., Lomov et al. and**
 250 **Chadha & Schwung. Record for each the exact sentence describing**
 251 **the data, or its absence.]**

252 3. Materials and Methods

253 3.1. Tennessee Eastman Process data

254 The TEP simulates a plant-wide reactor–separator–recycle system and
 255 includes 41 process measurements (XMEAS) and 11 manipulated variables
 256 (XMV), totaling 52 predictors [6]. The original problem statement specifies
 257 the plant but not a control structure; the simulations distributed as bench-
 258 mark data use the plant-wide control scheme of Lyman and Georgakis [21],
 259 so the recorded trajectories reflect both the process dynamics and the closed
 260 loops acting on them. The classification task contained 21 labels: class 0 rep-
 261 resented fault-free operation and classes 1–20 represented the predefined fault
 262 conditions. The CSV distribution used in this work is a format conversion
 263 of the expanded TEP simulation data of Rieth et al. [7, 15].

264 The four source files contained 15,330,000 observations and 55 columns:
 265 the 52 process variables plus the identifiers `faultNumber`, `simulationRun`,
 266 and `sample`. Development runs contained 500 observations each. Official
 267 test runs contained 960 observations each. Every class had 500 official test
 268 runs, yielding

$$21 \times 500 \times 960 = 10,080,000 \quad (1)$$

269 test observations. Each class therefore contributed 480,000 observations
 270 to the final test.

271 3.2. Fault-onset windows and labeling protocols

272 The expanded dataset is generated by simulating each condition for 25
 273 hours (development) or 48 hours (official test) with a sampling interval of

three minutes, and by injecting the fault after a fixed warm-up interval: one hour in faulty development runs and eight hours in faulty test runs [7, 16, 17]. In sample indices, the fault is active from sample 21 in development runs and from sample 161 in official test runs. Fault-free runs contain no injection.

Consequently, the labels distributed with the dataset are only conditionally correct. In a faulty development run, samples 1–20 (4.0% of the run) describe nominal operation while carrying a fault label; in a faulty official test run, samples 1–160 (16.7% of the run) do so. Two consequences follow directly and are, to our knowledge, seldom made explicit.

First, under an observation-level protocol with literal labels, the recall of every fault class is bounded above by

$$R_c^{\max} = \frac{960 - 160}{960} = 0.8333, \quad c = 1, \dots, 20, \quad (2)$$

whenever the classifier assigns pre-onset observations to some class other than c . No such ceiling applies to class 0, whose test runs contain no injection. Second, because the mislabeled fraction differs between development (4.0%) and test (16.7%), a metric computed on the two partitions is not measuring the same quantity, and part of any observed validation-to-test degradation is attributable to this asymmetry rather than to generalization.

We therefore report all headline metrics under two labeling protocols:

- **Literal labeling (L)**: every observation retains the label distributed with the dataset. This is the protocol implicitly adopted by most published observation-level TEP results and is retained here for comparability.
- **Post-onset labeling (P)**: pre-onset observations of faulty runs are excluded *at evaluation time*, so that each scored observation carries a label consistent with the physical state of the plant. The model is not refitted: protocol P rescores the predictions of the same frozen model that was fitted under protocol L, including its 4% of contaminated training observations. The comparison L versus P therefore isolates the labeling convention alone, and the reported post-onset figures are conservative, since a model additionally trained without the contaminated observations could only do better.

A third variant, in which pre-onset observations are excluded from fitting as well, was not evaluated here; it would confound the labeling effect with a change in the training distribution and is left for future work.

308 Protocol L is used for model selection, so that the frozen-model decision
 309 is unaffected by the present analysis. Protocol P is reported alongside it in
 310 Section 4.4. A third option, relabeling pre-onset observations as class 0, was
 311 not adopted because it alters the class prior of the normal condition and
 312 would confound the analysis of the ambiguity group.

313 3.3. Leakage-controlled partitioning and preprocessing

314 The original development runs were divided at the run level into training
 315 and validation partitions using a fixed random seed and a stratified 80/20
 316 split within each source and class. This produced 8,400 training runs (400
 317 per class), 2,100 validation runs (100 per class), and 10,500 official test runs
 318 (500 per class). No run was assigned to more than one partition.

319 All 52 predictors were standardized using

$$z_{ij} = \frac{x_{ij} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \quad (3)$$

320 where $\mu_j^{(\text{train})}$ and $\sigma_j^{(\text{train})}$ were estimated exclusively from the 4.2 million
 321 training observations. The fitted transformation was then applied unchanged
 322 to validation and test data. This procedure prevented information from the
 323 validation or test partitions from influencing preprocessing.

324 3.4. Candidate classifiers

325 Six model families were compared. Hyperparameter tuning evaluated
 326 three candidates per family, producing 18 configurations. The selected con-
 327 figurations were subsequently frozen:

- 328 • **Logistic regression:** $C = 1.0$, maximum 1,000 iterations;
- 329 • **Decision tree:** unrestricted depth and minimum leaf size 1;
- 330 • **Random forest:** 400 trees, unrestricted depth, 50% of features con-
 331 sidered per split, and minimum leaf size 1;
- 332 • **HistGradientBoosting:** learning rate 0.05, 350 boosting iterations,
 333 and at most 31 leaf nodes;
- 334 • **Linear SVM:** stochastic-gradient hinge-loss classifier with $\alpha = 10^{-5}$,
 335 maximum 2,000 iterations, and tolerance 10^{-3} ;

336 • **XGBoost**: 250 trees, maximum depth 6, learning rate 0.1, row sub-
 337 sampling 0.8, column subsampling 0.8, multiclass probabilistic objec-
 338 tive, and histogram tree construction.

339 The tuning stage ranked configurations by macro-F1 and used MCC as a
 340 secondary criterion. The official test partition was not accepted as an input
 341 by either the tuning or repeated-validation programs.

342 The linear SVM was implemented as a stochastic-gradient surrogate rather
 343 than as an exact margin solver, because quadratic-programming solvers were
 344 not tractable at this sample size. Its score should therefore be read as a lower
 345 bound on linear-margin performance, and the evidence for nonlinearity re-
 346 ported in Section 5.1 rests primarily on the multinomial logistic regression,
 347 which was optimized to convergence. Kernel SVMs were not evaluated for
 348 the same computational reason.

349 3.5. Repeated validation

350 After selecting the best configuration for each model family, variability
 351 was measured across seeds 2026–2030. For each seed, 40 complete runs per
 352 class were sampled from training and independently from validation. Thus,
 353 every repetition used 420,000 training and 420,000 validation observations
 354 while retaining the full 500-observation trajectories of all selected runs.

355 For each metric, the mean, sample standard deviation, and 95% confi-
 356 dence interval were calculated across the five repetitions. With $n = 5$, the
 357 interval was

$$\bar{x} \pm t_{0.975,4} \frac{s}{\sqrt{5}}. \quad (4)$$

358 Because only five seeds were used, these intervals quantify observed seed-
 359 to-seed variability but should not be interpreted as definitive evidence of
 360 pairwise statistical superiority. In particular, overlapping intervals for the
 361 two leading models were interpreted conservatively.

362 In addition to the interval estimates, the six model families were compared
 363 with the Friedman rank test over the five seeds treated as blocks, followed
 364 by the Nemenyi post-hoc procedure [19]. Pairwise comparisons used the
 365 Wilcoxon signed-rank test with Holm correction. The critical difference of
 366 the Nemenyi procedure was computed as $CD = q_\alpha \sqrt{k(k+1)/6N}$ with $k =$
 367 6 models and $N = 5$ blocks. Results are reported in Section 4.1. With
 368 five blocks these procedures have low power by construction, and they are

369 reported to document that the prespecified ranking rule was not substituted
370 for an inferential check, not to establish pairwise superiority.

371 Because a single frozen evaluation cannot express sampling uncertainty,
372 the official-test metrics were additionally accompanied by 95% intervals ob-
373 tained by resampling complete runs with replacement (1,000 bootstrap repli-
374 cates over the 10,500 test runs). The run, not the observation, is the resam-
375 pling unit, consistent with the argument that temporally adjacent observa-
376 tions are not independent replicates.

377 3.6. Frozen final evaluation

378 XGBoost candidate 1 was selected because it achieved the highest repeated-
379 validation macro-F1. The final seed was fixed at 2026 before test evaluation.
380 The final development set combined 40 complete runs per class from train-
381 ing and 40 from validation, totaling 840,000 observations. The frozen model
382 was then evaluated once on all 10,080,000 official test observations, without
383 subsampling or post-test adjustment.

384 Two earlier executions aborted during structural validation, before any
385 model fitting or metric computation, because of an incorrect run-length as-
386 sertion; only that assertion was changed, and the full incident log is available
387 in the repository.

388 The choice of 840,000 development observations, rather than the full
389 5,250,000 available, was imposed by the computational budget. To verify
390 that this choice does not materially affect the conclusions, a learning curve
391 was computed on the validation partition over $\{10, 20, 40, 80, 160\}$ complete
392 runs per class: **[TBD: learning-curve result and saturation point]**.

393 3.7. Performance measures and reproducibility

394 The reported measures were accuracy, balanced accuracy, precision, re-
395 call, class-specific F1, macro-F1, weighted-F1, MCC, confusion matrix, train-
396 ing time, inference time, inference time per observation, and serialized model
397 size. Zero divisions in class-specific measures were assigned zero. Data and
398 output integrity were monitored with SHA-256 hashes and manifests. The
399 final execution used Python 3.12.13, scikit-learn 1.9.0, and XGBoost 3.4.0 on
400 Linux.

Table 2: Repeated-validation performance over five seeds. Macro-F1 was the prespecified primary selection criterion.

Rank	Model	Macro-F1 (mean $\pm$ SD)	95% CI	MCC (mean $\pm$ SD)
1	XGBoost	0.7683 ± 0.0017	[0.7661, 0.7705]	0.7426 ± 0.0018
2	HistGradientBoosting	0.7643 ± 0.0034	[0.7601, 0.7686]	0.7391 ± 0.0045
3	Random Forest	0.7398 ± 0.0021	[0.7372, 0.7424]	0.7159 ± 0.0021
4	Decision Tree	0.6714 ± 0.0013	[0.6699, 0.6730]	0.6537 ± 0.0015
5	Logistic Regression	0.4685 ± 0.0074	[0.4592, 0.4777]	0.4567 ± 0.0073
6	Linear SVM	0.4438 ± 0.0099	[0.4314, 0.4561]	0.4377 ± 0.0109

4. Results

4.1. Model selection under repeated validation

Table 2 summarizes the five-seed validation results. XGBoost ranked first with macro-F1 of 0.7683 ± 0.0017 and MCC of 0.7426 ± 0.0018 . HistGradientBoosting was a close second, with macro-F1 of 0.7643 ± 0.0034 . Random forest achieved 0.7398, followed by decision tree at 0.6714. The two linear baselines performed substantially worse: logistic regression achieved 0.4685 and linear SVM achieved 0.4438.

The Friedman test over the five seeds rejected the hypothesis of equal performance among the six families ($\chi^2_5 = 24.54$, $p = 1.71 \times 10^{-4}$), with a Kendall coefficient of concordance of $W = 0.982$. The ordering is thus almost perfectly consistent from seed to seed: XGBoost obtained the best macro-F1 in four of the five repetitions, and the mean ranks were 1.2 (XGBoost), 1.8 (HistGradientBoosting), 3.0 (random forest), 4.0 (decision tree), 5.0 (logistic regression) and 6.0 (linear SVM).

Post-hoc resolution is nevertheless limited. With $N = 5$ blocks the Nemenyi critical difference is 3.372 mean ranks, so only three of the fifteen pairwise comparisons are individually significant, all of them between a tree ensemble and a linear baseline. The separation between XGBoost and HistGradientBoosting is 0.6 mean ranks, far below the critical difference. The Wilcoxon signed-rank test cannot help here either: with five paired observations its smallest attainable two-sided p -value is 0.0625, so no comparison can reach $\alpha = 0.05$ even under a perfect win record, and after Holm correction none does.

These two facts are complementary rather than contradictory. The high concordance shows that the ranking is not an artifact of seed choice; the wide critical difference shows that five repetitions cannot certify small dif-

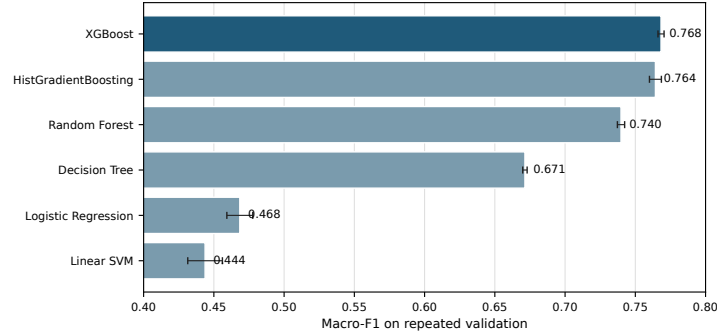


Figure 1: Repeated-validation macro-F1 for the six candidate algorithms. Error bars denote 95% confidence intervals across five seeds.

428 ferences. Both support the conservative reading adopted here: the ordering
 429 is reproducible, the gap between the two leading models is not resolvable at
 430 this sample size, and additional seeds would be required to settle it.

431 Figure 1 shows the ranking and its uncertainty. The confidence intervals
 432 of XGBoost and HistGradientBoosting overlapped. Accordingly, XGBoost
 433 was selected by the prespecified macro-F1 rule, but the result is not presented
 434 as proof of a statistically significant difference between the two models. The
 435 large separation between nonlinear ensembles and linear baselines indicates
 436 that nonlinear interactions are central to this diagnostic task.

437 4.2. Official test performance

438 After the protocol was frozen, the selected XGBoost model was fitted to
 439 840,000 development observations and evaluated on the complete official test
 440 set. Table 3 presents the final metrics under literal labeling. Accuracy and
 441 balanced accuracy were both 0.6754, macro-F1 and weighted-F1 were both
 442 0.7200, and MCC was 0.6621. The equality of accuracy and balanced accu-
 443 racy, and of macro-F1 and weighted-F1, follows from the exactly balanced
 444 class supports, and holds only under literal labeling; under the post-onset
 445 protocol of Section 4.4 the fault classes retain 400,000 observations while the
 446 fault-free class retains 480,000, so the two pairs of metrics separate. Macro-
 447 F1 exceeds accuracy because accuracy equals macro-recall under balanced
 448 supports, while precision exceeds recall in most classes; the harmonic mean
 449 of a high precision and a moderate recall is larger than the recall alone. The
 450 gap between macro-F1 and accuracy is therefore itself a signature of the
 451 systematic under-recall documented in Section 4.4.

Table 3: Frozen XGBoost performance on the official test set.

Metric	Value
Accuracy	0.6754
Balanced accuracy	0.6754
Macro-F1	0.7200
Weighted-F1	0.7200
MCC	0.6621
Training time (s)	170.34
Inference time (s)	56.57
Inference (ms/observation)	0.005612
Development observations	840,000
Test observations	10,080,000

Relative to repeated validation, macro-F1 decreased by 0.0483 and MCC by 0.0805. Three mechanisms could account for this difference: independent random states in the test simulations, the longer duration of test runs, and the larger fraction of mislabeled pre-onset observations. The third mechanism is both the largest and the only one that is purely a labeling artifact. Under literal labeling the attainable recall per fault class falls from $480/500 = 0.96$ in development runs to $800/960 = 0.8333$ in test runs, a ratio of 0.868, which is of the same order as the observed degradation. Section 4.4 shows that the gap does not merely shrink under post-onset labeling but changes sign: the post-onset test macro-F1 of 0.8054 exceeds the literal-label validation figure of 0.7683 by 0.0371.

That comparison alone would not be symmetric, because the validation figure was itself computed under literal labels and carries its own 4% pre-onset contamination. The validation partition was therefore rescored under both protocols as well. Applying the selected XGBoost configuration to all 1,050,000 validation observations gave a literal-label macro-F1 of 0.7706, which agrees with the five-seed repeated-validation figure of 0.7683 to within 0.0023 and confirms that this partition is a faithful reference. Excluding the 40,000 pre-onset observations, 3.81% of the partition, raised macro-F1 to 0.7953, accuracy from 0.7556 to 0.7843, and MCC from 0.7440 to 0.7739.

The correction on the validation partition is thus +0.0246 in macro-F1, against +0.0854 on the test partition, a ratio of 0.29 that tracks the ratio of contaminated fractions (3.81% against 15.87%). Transferring the measured validation correction to the repeated-validation reference gives a post-

476 onset validation macro-F1 of approximately 0.7929, against a post-onset test
 477 macro-F1 of 0.8054. Table 4 summarizes the decomposition.

Table 4: Validation-to-test difference in macro-F1 under both labeling protocols. The gap present under literal labeling does not survive a symmetric comparison.

Protocol	Validation	Test	Difference
Literal (L)	0.7683	0.7200	+0.0483
Post-onset (P)	0.7929	0.8054	−0.0125

478 Under a symmetric comparison the gap is not merely reduced but re-
 479 versed, and its magnitude falls to roughly a quarter of the literal-label value,
 480 within the range of the seed-to-seed variability of Table 2. We therefore
 481 conclude that the reported generalization gap is an artifact of the labeling
 482 convention rather than evidence of degradation on unseen trajectories, and
 483 that the literal-label figure of 0.7200 understates the discriminative capacity
 484 of the frozen model by 0.0854 in macro-F1.

485 Two caveats attach to this conclusion. The validation rescoring used the
 486 tuned configuration fitted to the full training partition, not the five per-seed
 487 refits of the repeated-validation procedure, so the transferred correction is an
 488 estimate of the effect rather than an exact recomputation; the close agree-
 489 ment between 0.7706 and 0.7683 bounds the error of that substitution. Fur-
 490 thermore, the pre-onset destinations differ between partitions: 61.1% of pre-
 491 onset validation observations were assigned to the group $\{0, 3, 9, 15\}$ against
 492 92.0% in the test partition, which is consistent with the development segment
 493 being only 20 samples long and overlapping the simulation transient.

494 4.3. Class-specific performance

495 Figure 2 and Table 5 demonstrate marked heterogeneity. Seventeen classes
 496 achieved F1 scores from 0.7429 to 0.9080. Classes 6 and 7 were strongest,
 497 with F1 of 0.9080 and 0.9075, respectively. Classes 1, 2, 5, and 14 also
 498 approached or exceeded 0.8958.

499 Four classes were persistently difficult: class 9 (F1=0.1635), class 15
 500 (0.1683), normal operation (class 0; 0.1718), and class 3 (0.2357). Class
 501 3 had the highest recall within this group (0.4218) but very low precision
 502 (0.1636), indicating systematic overprediction.

503 A pattern in the recall column deserves emphasis. Classes 6 and 7 reached
 504 recalls of 0.8333 and 0.8332, which coincide with the structural ceiling of

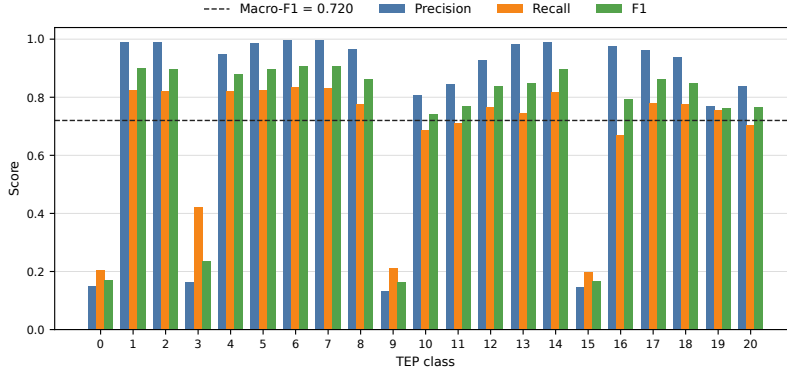


Figure 2: Class-wise precision, recall, and F1 on the complete official test partition. The dashed line indicates overall macro-F1. Class 0 is normal operation.

Eq. (2) to within one part in ten thousand. Classes 1, 2, 4, 5 and 14 lie between 0.8187 and 0.8243, that is, at 98–99% of that ceiling. The frozen model is therefore essentially saturated on these classes once the pre-onset segment is removed, and their reported F1 values of roughly 0.88–0.91 are limited by the labeling convention rather than by the classifier.

4.4. Effect of the fault-onset window

Table 6 compares the frozen model under the two labeling protocols of Section 3.2. No refitting, retuning or reselection was performed; the same frozen model and the same predictions are scored under two label conventions.

Excluding the 1,600,000 pre-onset observations, 15.87% of the partition, raised macro-F1 by 0.0854, accuracy by 0.1195 and MCC by 0.1231. No model was refitted and no hyperparameter was changed.

Two diagnostics establish that this is a labeling effect rather than a favourable choice of subset. First, the frozen model assigned 16.10% of the pre-onset observations to class 0 and 75.86% to classes 3, 9 or 15, that is, 92.0% to the group of conditions that are indistinguishable from nominal operation, and only 4.20% to the fault label they nominally carry. The model therefore identifies the pre-onset segment as a period in which nothing detectable is occurring, which is correct, and literal labeling scores this as error. Second, restricting the evaluation to post-onset observations raised the recall of the 17 well-separated classes to between 0.8015 and 0.9999, the upper value corresponding to essentially complete recovery of faults 6 and 7, while the

Table 5: Class-wise results for the frozen XGBoost model on the official test set. Every class had support of 480,000 observations.

Class	Precision	Recall	F1	Class	Precision	Recall	F1
0	0.1481	0.2045	0.1718	11	0.8447	0.7095	0.7712
1	0.9894	0.8243	0.8993	12	0.9271	0.7669	0.8395
2	0.9896	0.8209	0.8974	13	0.9828	0.7444	0.8471
3	0.1636	0.4218	0.2357	14	0.9889	0.8187	0.8958
4	0.9477	0.8222	0.8805	15	0.1465	0.1978	0.1683
5	0.9879	0.8234	0.8982	16	0.9756	0.6681	0.7931
6	0.9973	0.8333	0.9080	17	0.9624	0.7809	0.8622
7	0.9964	0.8332	0.9075	18	0.9367	0.7760	0.8488
8	0.9665	0.7779	0.8620	19	0.7701	0.7559	0.7630
9	0.1328	0.2126	0.1635	20	0.8391	0.7029	0.7650
10	0.8066	0.6885	0.7429				

Table 6: Frozen XGBoost under literal (L) and post-onset (P) labeling on the official test partition. Intervals are 95% run-level bootstrap intervals over 1,000 replicates.

Metric	Literal (L)	Post-onset (P)
Accuracy	0.6754 [0.6716, 0.6797]	0.7949 [0.7898, 0.8006]
Balanced accuracy	0.6754 [0.6748, 0.6760]	0.8006 [0.7998, 0.8013]
Macro-F1	0.7200 [0.7192, 0.7208]	0.8054 [0.8044, 0.8063]
MCC	0.6621 [0.6579, 0.6666]	0.7852 [0.7798, 0.7910]
Observations	10,080,000	8,480,000

ambiguity group improved only marginally (F1 of 0.2217, 0.3238, 0.2100 and 0.2106 for classes 0, 3, 9 and 15, against 0.1718, 0.2357, 0.1635 and 0.1683 under literal labels). The ambiguity group is therefore not a labeling artifact; the ceiling on the remaining seventeen classes is.

The two phenomena are nevertheless connected. Class 3 was predicted 1,237,562 times against a true support of 480,000, an excess of 757,562 observations. A substantial part of that excess consists of pre-onset segments belonging to other faults, which the model routes to the weakly observable conditions because, at those instants, nothing has yet happened. The over-prediction of class 3 is thus partly a consequence of the labeling convention and partly a consequence of genuine overlap, and only the post-onset analysis separates the two.

Table 7: Effect of post-onset labeling across model families. The last column reports the relative accuracy gain $\Delta A/A_P$, whose structural upper bound is $1 - 8,480,000/10,080,000 = 0.1587$.

Model	Macro-F1 (L)	Macro-F1 (P)	$\Delta A/A_P$
Decision tree	0.6330	0.7167	0.148
Extra trees	0.6965	0.7786	0.148
Random forest	0.7081	0.7912	0.149
HistGradientBoosting	0.7128	0.7972	0.149
XGBoost [†]	0.7200	0.8054	0.150
Logistic regression	0.4283	0.4698	0.142
Linear SVM	0.4207	0.4655	0.142

[†]Frozen model of Table 3, complete test partition.

540 This distinction matters for comparability. As documented in Section 2.5,
541 published observation-level results on the expanded TEP dataset do not con-
542 sistently state whether pre-onset samples were retained, discarded, or rela-
543 beled, yet the three choices are separated by several points of macro-F1 for an
544 identical model. Two studies reporting different scores may therefore differ
545 in labeling convention rather than in method. We recommend that the treat-
546 ment of the onset window be reported explicitly as part of the experimental
547 protocol, in the same way that the unit of partitioning is reported.

548 4.5. Generality of the labeling effect across model families

549 If the effect of Section 4.4 is a property of the labeling convention rather
550 than of the selected classifier, it should appear in every model evaluated on
551 the same partition. To test this, an auxiliary experiment was rescored un-
552 der both protocols. That experiment predates the frozen benchmark, uses
553 a per-class subsample of the test partition, and includes an extremely ran-
554 domized trees model in place of XGBoost; its absolute levels are therefore
555 not comparable with Table 3, and only the differences between protocols are
556 interpreted. No model was refitted.

557 The five tree-based models gained between 0.0821 and 0.0854 in macro-
558 F1, a spread of 0.003 across estimators whose capacities differ by orders of
559 magnitude, from a single unpruned tree to a regularized boosting ensemble.
560 No overfitting mechanism produces that uniformity, because the degradation
561 associated with overfitting scales with model capacity.

562 The relative accuracy gain explains the pattern quantitatively. If a clas-
 563 sifier assigns pre-onset observations to any class other than the labeled fault,
 564 its literal accuracy is its post-onset accuracy scaled by the retained fraction
 565 of the partition, so that $\Delta A/A_P$ is bounded above by 0.1587 independently
 566 of the model. The seven models fall between 0.142 and 0.150, the residual
 567 being the small fraction of pre-onset observations that match the literal la-
 568 bel by chance. The linear baselines gain less in absolute terms, 0.0415 and
 569 0.0448, simply because they start lower; in relative terms they are affected
 570 almost identically. They also route far fewer pre-onset observations to the
 571 group $\{0, 3, 9, 15\}$, 20.6% and 41.9% against 65.7–70.2% for the ensembles,
 572 and are penalized in the same proportion regardless.

573 The consequence is that the effect is not a property of any estimator.
 574 Any observation-level result reported on this dataset under literal labels un-
 575 derstates the post-onset discriminative capacity of the model by a factor
 576 predictable from its own accuracy. Comparisons between published results
 577 are correspondingly affected whenever the treatment of the onset window
 578 differs or is unstated.

579 *4.6. Structure of the remaining errors*

580 The normalized confusion matrix in Fig. 3 shows that errors were not
 581 uniformly distributed. They concentrated in the four-class group $\{0, 3, 9, 15\}$.
 582 Among true class-9 observations, 32.39% were predicted as class 3, 20.24%
 583 as class 0, and 18.08% as class 15. Normal observations were assigned to
 584 classes 3, 9, and 15 in 32.02%, 21.51%, and 18.24% of cases, respectively. For
 585 class 15, the corresponding proportions were 30.96%, 20.53%, and 20.03%
 586 for predicted classes 3, 9, and 0.

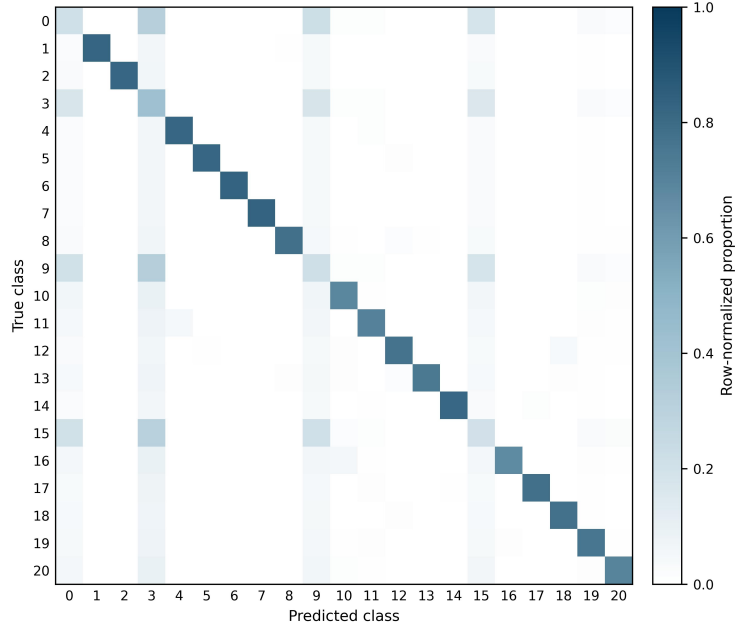


Figure 3: Row-normalized confusion matrix for the frozen XGBoost model on the official test partition.

Class 3 was predicted 1,237,562 times despite having 480,000 true observations. This excess explains its low precision and demonstrates that the model used class 3 as a frequent destination for ambiguous trajectories. The persistence of the same difficult group during repeated validation and official testing suggests an intrinsic overlap under an observation-level representation rather than an isolated sampling artifact.

4.7. Computational performance

Table 8 reports the cost of all six families under identical conditions during repeated validation, which is required to answer the performance–cost question and cannot be inferred from the selected model alone. Training times are reported as means over the five seeds and are not directly comparable with the 170.34s of Table 3, which was measured during the final evaluation on twice as many observations. The degree of parallelism was an invocation-time argument of each stage rather than a fixed setting, and it was not recorded in the run manifests, so the absolute times of the two stages reflect different thread counts on a 24-core host. The comparison between

Table 8: Performance and computational cost of the six families under repeated validation (mean over five seeds, 420,000 training observations per repetition). Times measured on the environment of Section 3.7.

Model	Macro-F1	MCC	Train (s)	Inference (μ s/obs)	Size (MB)
XGBoost	0.7683	0.7426	864.7	7.03	18.9
HistGradientBoosting	0.7643	0.7391	73.5	29.40	16.9
Random Forest	0.7398	0.7159	477.3	20.29	8703.8
Decision Tree	0.6714	0.6537	54.0	0.24	28.8
Logistic Regression	0.4685	0.4567	65.4	0.15	0.005
Linear SVM	0.4438	0.4377	4.1	0.16	0.006

families within Table 8 is unaffected, because those six runs shared a single invocation, but the cross-stage comparison is not meaningful and no conclusion is drawn from it. Recording the degree of parallelism alongside the timing is a reproducibility requirement that this benchmark did not initially meet, and it has been added to the released code.

Final XGBoost training required 170.34 seconds. Prediction over 10.08 million observations required 56.57 seconds, corresponding to 0.005612 ms per observation or approximately 178,000 observations per second. The serialized final model occupied approximately 6.8 MB. These values indicate that the selected classifier is computationally feasible for high-throughput offline analysis and potentially for online scoring, although end-to-end deployment latency must also include data acquisition, preprocessing, communication, and decision logic.

5. Discussion

5.1. Nonlinearity and model ranking

The ranking provides strong empirical evidence that the TEP classification boundary is not adequately represented by the evaluated linear models. XGBoost, HistGradientBoosting, and random forest substantially outperformed logistic regression and linear SVM even though all models received the same standardized predictors and run-level partitions. Tree ensembles can represent threshold effects and high-order interactions without requiring an explicit feature expansion, which is consistent with the nonlinear and feedback-driven nature of the TEP.

The difference between XGBoost and HistGradientBoosting was small: 0.0040 in mean macro-F1, with overlapping 95% confidence intervals. Thus,

the defensible conclusion is that XGBoost won under the prespecified ranking rule, not that it was conclusively superior under all resampling regimes. Table 8 makes the practical consequence explicit, and it does not favour a single model. HistGradientBoosting trained 11.8 times faster than XGBoost (73.5s against 864.7s) for a deficit of 0.0040 in macro-F1 with overlapping intervals, so it dominates whenever models are retrained frequently or the training budget is constrained. The ordering reverses at inference time, where XGBoost required $7.03\ \mu\text{s}$ per observation against $29.40\ \mu\text{s}$, a factor of 4.2 in favour of XGBoost. Neither model dominates the other: the choice follows from whether the deployment is retraining-bound or scoring-bound, and both conclusions are invisible if only the selected model is profiled.

Model size adds a third and largely independent constraint. The random forest required approximately 8.7 GB, some 460 times the XGBoost artifact, for 0.0285 less macro-F1. Whatever its statistical merits, it is not a candidate for embedded or edge deployment, and this would not be apparent from predictive metrics alone. The two linear baselines occupied roughly 5 kB and inferred in $0.15\ \mu\text{s}$ per observation, which quantifies precisely what is being paid for the nonlinear gain of about 0.30 in macro-F1.

5.2. *Why global performance is insufficient*

The official macro-F1 of 0.7200 summarizes useful performance, but it does not describe a uniformly reliable 21-class diagnostic system. If only macro-F1 or accuracy had been reported, the strong results for 17 classes would obscure the near-failure of normal operation and faults 3, 9, and 15. In operational terms, low precision for normal operation and extensive exchange between normal and faulty labels can increase false alarms and missed interventions.

This result illustrates why class-level precision, recall, F1, and confusion structure should accompany aggregate measures in industrial diagnosis. It also supports a distinction between *benchmark discrimination* and *deployment readiness*. The present model is a strong benchmark classifier for most faults, but additional safeguards are necessary before it could support autonomous decisions.

5.3. *Interpretation of the ambiguity group*

The identity of the difficult group is not a new finding, and it would be misleading to present it as one. Faults 3, 9 and 15 have been reported as

663 weakly detectable since the earliest systematic studies of the TEP: moni-
664 toring statistics based on PCA, discriminant PLS and Fisher discriminant
665 analysis report missed-detection rates close to those of nominal operation
666 for exactly these conditions [5, 4], and later comparative studies reproduce
667 the pattern across a wide range of data-driven statistics [18]. The contribu-
668 tion here is the scale and the protocol: the same group is recovered under
669 run-level partitioning, on 10.08 million independent test observations, and it
670 persists after the labeling artifact of Section 4.4 is removed. This strengthens
671 the interpretation that the overlap is a property of the measured variables
672 rather than of any particular estimator.

673 The mechanism, however, constrains which remedies can work. If these
674 disturbances leave no persistent signature in the 52 measured variables, then
675 temporal aggregation over a run reduces the variance of an estimate that is
676 already centered on an ambiguous region: it will sharpen the decision without
677 moving it toward the correct class. Run-level aggregation should therefore
678 be expected to help the 17 separable classes, whose pre-onset transients are
679 transient, far more than the group $\{0, 3, 9, 15\}$, and this asymmetry is itself a
680 testable prediction. A hierarchical design that isolates the ambiguous group
681 and applies a discriminator with explicit memory to classes 0, 3, 9 and 15
682 is the more promising direction, but its plausibility rests on whether any
683 signature exists at all; establishing this is a question of observability, not
684 of classifier capacity, and it is better addressed with residual-based or first-
685 principles analysis than with additional supervised tuning.

686 Both proposals arise from test-set diagnosis and must therefore be eval-
687 uated under a newly defined development/test protocol; the present official
688 test must not be reused for tuning them.

689 *5.4. Threats to validity*

690 Several limitations delimit the conclusions. First, TEP is a high-fidelity
691 simulation benchmark, not a physical plant. Results may not transfer directly
692 to sensor drift, maintenance changes, unmodeled disturbances, communica-
693 tion losses, or evolving operating modes in real facilities. Second, the bench-
694 mark evaluated pointwise multiclass diagnosis and did not measure detection
695 delay, time-to-alarm, false alarms per run, or sustained alarm logic. Third,
696 five seeds provide a useful but small basis for uncertainty estimation. The
697 resulting t intervals characterize the implemented experiment rather than all
698 possible data partitions.

Fourth, hyperparameter search was intentionally limited to three candidates per model family to preserve computational feasibility and transparency. A larger search might improve individual algorithms, particularly the linear baselines or random forest. Fifth, the final model was fitted to 840,000 of the 5,250,000 available development observations, and model selection was carried out at 420,000; a ranking established at one training scale need not be invariant to scale, and the learning curve reported in Section 4.1 bounds but does not eliminate this concern. Sixth, the linear SVM was a stochastic-gradient surrogate and its result should not be read as the best attainable linear-margin performance. Seventh, the degree of parallelism was not recorded in the run manifests, so the training times of Table 8 support ordinal comparison between families, which shared one invocation, but not comparison against the timings of other stages. Eighth, interpretation focused on output behavior and did not include feature attribution or causal process analysis. Finally, the dataset had balanced class support, whereas real industrial faults are rare and highly imbalanced. Metrics, thresholds, and alarm costs would require recalibration under field prevalence.

6. Conclusion

This work presented a reproducible classical-machine-learning benchmark for 21-class fault diagnosis in the Tennessee Eastman Process. Run-level splitting, train-only standardization, separate tuning and validation, five-seed repetition, and a single frozen official test evaluation were used to reduce leakage and post-selection bias.

XGBoost achieved the highest repeated-validation macro-F1 (0.7683 ± 0.0017) and produced an official test macro-F1 of 0.7200, accuracy of 0.6754, and MCC of 0.6621 over 10.08 million observations under literal labeling, rising to a macro-F1 of 0.8054, an accuracy of 0.7949 and an MCC of 0.7852 once the pre-onset segment of each faulty run is excluded, with no refitting of any kind. The validation partition, rescored under the same protocol, moved from 0.7706 to 0.7953, so that the apparent generalization gap of +0.0483 in macro-F1 becomes -0.0125 once the comparison is made symmetric. Seventeen classes were diagnosed with F1 between 0.7429 and 0.9080, and several of them were shown to sit at the structural recall ceiling imposed by the labeling convention rather than at the limit of the classifier. Normal operation and faults 3, 9, and 15 remained strongly confounded, in agreement with

734 two decades of TEP monitoring literature, and this confounding survives the
735 removal of the labeling artifact.

736 Three conclusions follow. Nonlinear tree ensembles provide effective di-
737 agnosis for most TEP conditions. Aggregate scores overstate practical reli-
738 ability when a small group of classes dominates the remaining errors. And
739 a benchmark result on the expanded TEP dataset is not interpretable, nor
740 comparable across studies, unless the treatment of the fault-onset window
741 is stated explicitly. A single frozen model differs by 0.085 in macro-F1 and
742 0.123 in MCC between two defensible labeling conventions, which is larger
743 than most of the improvements reported between competing architectures on
744 this benchmark. The effect is moreover common to every model family tested,
745 with a relative accuracy gain between 0.142 and 0.150 across seven classifiers,
746 so it displaces published results without reordering them only when all stud-
747 ies happen to adopt the same convention. We therefore recommend that the
748 treatment of the onset window be stated as part of the experimental protocol
749 in any observation-level study using these data.

750 Future work should investigate temporal aggregation, hierarchical diag-
751 nosis, probability calibration, detection delay, and cost-sensitive alarm rules
752 under a new untouched test protocol. External validation with physical in-
753 dustrial data will be necessary before deployment-oriented conclusions can
754 be drawn.

755 Data and Code Availability

756 The expanded Tennessee Eastman simulation data are publicly available
757 from Harvard Dataverse [7]. The CSV conversion used in this study is avail-
758 able through Kaggle [15]. All code, configuration files, run-level split mani-
759 fests, frozen hyperparameters, five-seed validation outputs, final predictions,
760 class-level metrics, confusion matrices, SHA-256 data hashes, and the en-
761 vironment specification are deposited at an anonymized repository for peer
762 review, **[TBD: anonymous DOI]**; the non-anonymized repository is cited
763 in the accepted version. The analysis of the fault-onset window can be re-
764 produced from the deposited predictions without refitting any model.

765 Conflict of Interest

766 The author declares no conflict of interest.

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